Physics 468/668: Statistical Physics **Spring 2019**

Lecture 12: Examples

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12.1 Ideal gas

For the canonical ensemble, we have the partition function as

$$Q_N(V,T) = \frac{[Q_1(V,T)]^N}{N!} \tag{12.1}$$

where $Q_1(V,T)$ is the partition function of a single molecule. Since the particles are non-localized they can be anywhere, the function Q_1 could be written as

$$Q_1(V,T) = Vf(T) \tag{12.2}$$

where $f(T)$ is a function of temperature alone. We thus obtain for the grand partition of the gas,

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} Z^{N_r} Q_{N_r}(V,T) = \sum_{N_r=0}^{\infty} \frac{[zVf(T)]^{N_r}}{N_r!} = \exp[zVf(T)] \tag{12.3}$$

which gives

$$q(z,V,T) = zVf(T) \tag{12.4}$$

We can then derive the following quantities,

$$\begin{aligned} P &= zkTf(T) \\ N &= zVf(T) \\ U &= zVkT^2 f'(T) \\ A &= NkT\ln z - zVkTf(T) \\ S &= NkT\ln z + zVk[Tf'(T) + f(T)] \end{aligned} \tag{12.5}$$

Again, we can also derive $PV = NkT$ from the above results. Considering N and U, we get

$$U = NkT^2 f'(T)/f(T) \tag{12.6}$$

which gives,

$$C_V = Nk\frac{2Tf(T)f'(T) + T^2[f(T)f''(T) - [f'(T)]^2]}{[f(T)]^2} \tag{12.7}$$

In simple cases, the function $f(T)$ turns out to be directly proportional to a certain power of T. Assuming $f(T) \propto T^n$, equations become

$$U = n(NkT) \tag{12.8}$$

$$C_V = n(Nk) \tag{12.9}$$

Accordingly, the pressure in such cases is directly proportional to the energy density of the gas.

Finally, eliminating z between N, A, S, we obtain A and S as functions of N, V, T.

12.2 Harmonic oscillator

The microscopic entities constituting the system are mutually indistinguishable. The partition function $Q_N(V,T)$ of such a system is:

$$Q_N(V,T) = [Q_1(V,T)]^N \tag{12.10}$$

At the same time, the single partition function $Q_1(V,T)$ is independent of the volume, so we write

$$Q_1(V,T) = \phi(T) \tag{12.11}$$

where $\phi(T)$ is a function of temperature alone. We then obtain the grand partition function of the system,

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} [z\phi(T)]^{N_r} = [1 - z\phi(T)]^{-1} \tag{12.12}$$

The thermodynamics of the system follows straightforwardly,

$$\begin{aligned} P &= \frac{kT}{V} q(z,T) = -\frac{kT}{V} \ln[1 - z\phi(T)] \\ N &= \frac{z\phi(T)}{1 - z\phi(T)} \\ U &= \frac{zkT^2\phi(T)}{1 - z\phi(T)} \\ A &= NkT \ln z + kT \ln[1 - z\phi(T)] \\ S &= -Nk \ln z - k \ln[1 - z\phi(T)] + \frac{zkT\phi'(T)}{1 - z\phi(T)} \end{aligned} \tag{12.13}$$

From N, we get

$$z\phi(T) = \frac{N}{N+1} \approx 1 - \frac{1}{N} \quad (N \gg 1) \tag{12.14}$$

It follows that

$$1 - z\phi(T) = \frac{1}{N+1} \approx \frac{1}{N} \tag{12.15}$$

So

$$U/N = kT^2\phi'(T)/\phi(T) \tag{12.16}$$

$$A/N = -kT \ln \phi(T) + O\left(\frac{\ln N}{N}\right) \tag{12.17}$$

substituting

$$\phi(T) = [2 \sinh(\hbar\omega/2kT)]^{-1} \tag{12.18}$$

into these formulas, we obtain the results of quantum mechanical 1D harmonic oscillators.

substituting

$$\phi(T) = kT/\hbar\omega \tag{12.19}$$

we get the results for classical 1D harmonic oscillators.

12.3 Density and energy fluctuation

How will the thermodynamics vary from different ensembles? The true answer turns out that the differences might not be that much, if the fluctuation of these variables is small (practically negligible). Therefore, it is useful to evaluate the fluctuations.

Recalling that

$$\overline{N} = \frac{\sum_{r,s} N_r e^{-\alpha N_r - \beta E_s}}{\sum_{r,s} e^{-\alpha N_r - \beta E_s}} \tag{12.20}$$

It follows that,

$$\left(\frac{\partial \overline{N}}{\partial \alpha}\right)_{\beta, E_s} = -\overline{N^2} + \overline{N}^2 \tag{12.21}$$

Thus,

$$\overline{(\Delta N)^2} = \overline{N^2} - \overline{N}^2 = \left(\frac{\partial \overline{N}}{\partial \alpha}\right)_{T,V} = kT\left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V} \tag{12.22}$$

The mean square fluctuation in the particle density $n = (N/V)$

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{\overline{(\Delta N)^2}}{\overline{N}^2} = \frac{kT}{\overline{N}^2}\left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V} \tag{12.23}$$

In terms of the variable $v = V/\overline{N}$, we write

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{kTv^2}{V^2}\left(\frac{\partial (V/v)}{\partial \mu}\right)_{T,V} = \frac{kT}{V}\left(\frac{\partial v}{\partial \mu}\right)_{T} \tag{12.24}$$

Recall the thermodynamic relation

$$d\mu = vdP - sdT \tag{12.25}$$

we get

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{kT}{V}\frac{1}{v}\left(\frac{\partial v}{\partial \mu}\right)_{T} = \frac{kT}{V}\kappa_T \tag{12.26}$$

where κ_T is the isothermal compressibility of the system. Thus, the rms of particle density is $O(N^{-1/2})$ and hence negligible. However, there are some exceptions. When it comes to the phase transitions, the compressibility of a given system can be very large!

We shall now examine the fluctuation in the energy of the system.

$$\overline{(\Delta E)^2} = \bar{E^2} - \bar{E}^2 = -\left(\frac{\partial \bar{E}}{\partial \beta}\right)_{z,V} = kT^2\left(\frac{\partial U}{\partial T}\right)_{z,V} \tag{12.27}$$

To make it more comprehensible,

$$\left(\frac{\partial U}{\partial T}\right)_{N,V} + \left(\frac{\partial U}{\partial N}\right)_{T,V}\left(\frac{\partial N}{\partial T}\right)_{z,V} \tag{12.28}$$

now, in view of the fact that

$$N = -\left(\frac{\partial}{\partial \alpha}\ln \mathcal{Q}\right)_{\beta,V}, \quad U = -\left(\frac{\partial}{\partial \beta}\ln \mathcal{Q}\right)_{\alpha,V} \tag{12.29}$$

we have

$$\left(\frac{\partial N}{\partial \beta}\right)_{\alpha,V} = \left(\frac{\partial N}{\partial \alpha}\right)_{\beta,V} \tag{12.30}$$

and

$$\left(\frac{\partial N}{\partial T}\right)_{z,V} = \frac{1}{T}\left(\frac{\partial U}{\partial \mu}\right)_{T,V} \tag{12.31}$$

substituting it, we get

$$\overline{(\Delta E)^2} = kT^2 C_V + kT\left(\frac{\partial U}{\partial N}\right)_{T,V}\left(\frac{\partial U}{\partial \mu}\right)_{T,V} \tag{12.32}$$

Finally,

$$\overline{(\Delta E)^2} = \overline{(\Delta E)^2}_{\text{canonical}} + \left[\frac{\partial U}{\partial N}\right]^2_{T,V} \overline{(\Delta N)^2} \tag{12.33}$$

It ends up with the picture that the value of canonical ensemble plus a contribution due to particle fluctuation.